Both include and exclude can be made up of explicit file names or you can use patterns with ? to match a single character, * to match any characters except directory separators, and ** to match all subdirectories. Our project is currently excluding specs (**/*.spec.ts), and three other directories that might have files we don't want to deal with. A file matched by include can be blocked by exclude, but a file in the files list is compiled even if it matches an exclude pattern.

There are quite a few compiler options.[77] In general, the options either govern the behavior of the compiler in judging what code is legal, or they change the nature of the JavaScript code generated by the compiler.

Note that in our particular setup, the compiler options are handled by tsc_watch, and the generating code is handled by esbuild. In general, esbuild only pays attention to the subset of configuration options having to do with output code; see https://esbuild.github.io/content-types/#tsconfig-json for up-to-date details.

Let's look at the options already specified in our project.

Handling input we have the experimentalDecorators property, which allows us to use the new decorator JavaScript syntax. We've got the lib property, which covers what versions of the standard JavaScript library will be allowed in TypeScript. We're using the es6 version, which in addition to syntax, has some implications for what methods exist on basic types like strings. We're also using the dom library, which gives us DOM-related classes like HTMLElement. There are a number of other options, including newer and more experimental ES versions.

A lot of properties govern module lookup. The module property, which we set to es6, governs what syntax TypeScript will use to specify modules, and the moduleResolution property governs the algorithm TypeScript uses to determine which module you are pointing to with a declaration (you're unlikely to change that one, I think). The baseUrl specifies the top-level directory for module declarations that are not relative to the current directory, and paths indicates other places to look for relative module declarations.

Specifying output, we have the "declaration": false, which tells the compiler not to